Ziwei Ma

Associate Professor
Department of Mathematics
The University of Tennessee at Chattanooga
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Education

New Mexico State University — **Las Cruces, NM** — **Jan 2014 – May 2020**

- **Ph.D. in Mathematics (Concentration: Mathematical Statistics)**
  Advisor: Prof. Tonghui Wang *Thesis title: "Estimation of Parameters and Decomposition of Quadratic Forms Under Skew-Normal Settings"*
- **M.S. in Mathematics**

Northwest University — **Xi'an, China** — **Aug 2000 – Jul 2007**

- **M.S. in Mathematics**
- **B.S. in Mathematics**

Professional Experience

Associate Professor	**University of Tennessee at Chattanooga**	**Aug 2026 – Present**
Assistant Professor	**University of Tennessee at Chattanooga**	**Aug 2020 – July 2026**
Biostatistician (part time)	**University of Tennessee College of Medicine - Chattanooga**	**Aug 2021 - Present**
Teaching/Research Assistant	**New Mexico State University**	**Jan 2014 - May 2020**

Research Interests

Statistics

- Linear Model, Skew Normal Distribution, Inferential Models, Clinic Trials

Applied Math/Stats

- Machine Learning in Engineering, Applications of Stochastic Differentiation Equations in Biology and Medical Science

Publications

Journal/Book Chapter/Conference/Preprint, Full list in Google scholar

2026

1. **[C] Ziwei Ma**, Yingfeng Wang, (2026) Information-Geometric Expressivity Diagnostics for Quantum Molecular Autoencoders via Fisher Effective Rank: A Theoretical Study, AAAI Fall Symposium Series 2026 (Accepted)

2. [**C**] Pairitz M, Buerster K, Rippy M, Stewart A, Benton M, Patel H, Gates H, Wise L, **Ziwei Ma**, Hunt D, Bell C. Prehospital Shock Index is Associated with Admission Hypocalcemia Following Traumatic Injury. Western Trauma Association 56th Annual Meeting. Earl G. Young Resident Paper Competition for Clinical Research. Whistler, BC. (Submitted, Pending Acceptance)

2025

3. [**J**] Raine, J. D., Rippy, M. G., Miles, M. V. P., **Ziwei Ma**, Carter, B. L., and Bell, C. (2025). Re-Evaluating the Necessity of Empiric Anti-Fungal Coverage in Gastroduodenal Perforations. The American Surgeon™, 00031348251367025.

2024

4. [**J**] Xiaonan Zhu, Zheng Wei, Tonghui Wang, S. T. Choy, and **Ziwei Ma**. "An expectation conditional maximization algorithm for the skew-normal based stochastic frontier model." Computational Statistics 39, no. 3 (2024): 1539-1558.
5. [**C**] N. Brott, G. Bose, I. Hembree, **Ziwei Ma**, R. Shah, and P. Koo. "Effect of Heterozygote Alpha-1-antitrypsin Gene on the Development of Pulmonary Fibrosis and Mortality (abstract)". Am J Respir Crit Care Med 2024;209:A2112.

2023

5. [**B**] **Ziwei Ma**, Tonghui Wang, ST Boris Choy, Zheng Wei, and Xiaonan Zhu. "Extending the A Priori Procedure for Estimating Location Parameter Under Multivariate Skew Normal Settings." In Optimal Transport Statistics for Economics and Related Topics, pp. 107-116. Cham: Springer Nature Switzerland, 2023.

2022

7. [**P**] Yifan Hao, Huiping Cao, K. Selcuk Candan, Jiefei Liu, Huiying Chen, and **Ziwei Ma**. "Class-Specific Attention (CSA) for Time-Series Classification." arXiv preprint arXiv:2211.10609 (2022).
8. [**C**] Gresham, Brandon, Juan Torres, Jonathan Britton, **Ziwei Ma**, Anita B. Parada, Michelle L. Gutierrez, Mark Lawrence, and Wei Tang. "High-dimensional Time-series Gait Analysis using a Full-body Wireless Wearable Motion Sensing System and Convolutional Neural Network." In 2022 IEEE Biomedical Circuits and Systems Conference (BioCAS), pp. 389-393. IEEE, 2022.
9. [**J**] **Ziwei Ma**, Ying-Ju Chen, Tonghui Wang, and Jing Liu. "Inferences on location parameter in multivariate skew-normal family with unknown scale parameter." Communications in Statistics-Simulation and Computation 51, no. 9 (2022): 5465-5481.
10. [**J**] **Ziwei Ma**, Tonghui Wang, Zheng Wei, and Xiaonan Zhu. "Inferences on location parameters based on independent multivariate skew normal distributions." Asian Journal of Economics and Banking 6, no. 2 (2022): 270-281.

2021

11. [**B**] **Ziwei Ma**, Wang, Tonghui, Li, Baokun, Xiaonan Zhu and yuede Ma. "The Decomposition of Quadratic Forms Under Matrix Variate Skew-Normal Distribution," In Behavioral Predictive Modeling in Economics (pp. 181-194). Springer, Cham. (2021)
12. [**B**] Li, Baokun, **Ziwei Ma**, and Tonghui Wang. "Find Trade Patterns in China's Stock Markets Using Data Mining Agents." Behavioral Predictive Modeling in Economics (2021): 171-179.

2020

13. [**J**] Hanwan Jiang, Hanyu Zhan, **Ziwei Ma**, and Ruinian Jiang. "Comparative Study of Three-Dimensional Stress and Crack Imaging in Concrete by Application of Inverse Algorithms to Coda Wave Measurements." Sensors 20, no. 17 (2020): 4899.
14. [**J**] Lei Mu, Zhe Jia, **Ziwei Ma**, Fuhui Shen, Yuekuo Sun, and Yong Zang. "A theoretical prediction framework for the construction of a fracture forming limit curve accounting for fracture pattern transition." International Journal of Plasticity 129 (2020): 102706.

15. [**J**] **Ziwei Ma**, Ben Niu, Tuan Anh Phan, Anne Line Stensjøen, Chibawanye Ene, Timothy Woodiwiss, Tonghui Wang, Philip K. Maini, Eric C. Holland, and Jianjun Paul Tian. "Stochastic growth pattern of untreated human glioblastomas predicts the survival time for patients." Scientific reports 10, no. 1 (2020): 6642.

2019

16. [**J**] Xiaochen Tang, **Ziwei Ma**, Qisong Hu, and Wei Tang. "A real-time arrhythmia heartbeats classification algorithm using parallel delta modulations and rotated linear-kernel support vector machines." IEEE Transactions on Biomedical Engineering 67, no. 4 (2019): 978-986.
17. [**J**] **Ziwei Ma**, Jinlei Hao, and Tonghui Wang. "The plausibility regions for shape parameter under multivariate skew-normal settings." International Journal of Intelligent Technologies & Applied Statistics 12, no. 1 (2019): 31-47.

2018

18. [**B**] **Ziwei Ma**, Ying-Ju Chen, Tonghui Wang, and Wuzhen Peng. "The inference on the location parameters under multivariate skew normal settings." In International Econometric Conference of Vietnam, pp. 146-162. Cham: Springer International Publishing, 2018.

2017

19. [**B**] **Ziwei Ma**, Xiaonan Zhu, Tonghui Wang, and Kittawit Autchariyapanitkul. "Joint plausibility regions for parameters of skew normal family." In International Conference of the Thailand Econometrics Society, pp. 233-245. Cham: Springer International Publishing, 2017.
20. [**B**] **Ziwei Ma**, Weizhong Tian, Baokun Li, and Tonghui Wang. "The decomposition of quadratic forms under skew normal settings." In International conference of the Thailand econometrics society, pp. 222-232. Cham: Springer International Publishing, 2017.
21. [**B**] Xiaonan Zhu, **Ziwei Ma**, Tonghui Wang, and Teerawut Teetranont. "Plausibility regions on the skewness parameter of skew normal distributions based on inferential models." Robustness in econometrics (2017): 267-286.

2014

22. [**J**] Junli Du, Zhifa Yuan, **Ziwei Ma**, Jiuzhou Song, Xiaoli Xie, and Yulin Chen. "KEGG-PATH: Kyoto encyclopedia of genes and genomes-based pathway analysis using a path analysis model." Molecular BioSystems 10, no. 9 (2014): 2441-2447.

Grants

- (**NSF, PI**) SCH: Collaborative Research: Rehabilitation Evaluation of Neurologic Motor Disorders using Wearable Sensors (**$ 189,931, Oct. 2026 - Sept. 2030**)
- (**UTCOM-C/Erlanger/UTC, PI**) Prevalence and Disease Characteristics of Alpha-1 Antitrypsin Deficiency (AAT) in a Southeastern United States Patient Population and its Association with Chronic Obstructive Pulmonary Disease (COPD), Pulmonary fibrosis, Hospital Readmission, and Mortality (**$ 25,000, Jan 2023 - June 2024**)
- (**NSF, Co-PI**) PIPP Phase I: Develop and Evaluate Computational Frameworks to Predict and Prevent Future Coronavirus Pandemics (**$ 1,000,000, Aug 2022 - Jan 2026**)
- (**SimCenter, Co-PI**) Integrating google trends analytic into geographically weighted model of vaccine hesitancy (**$ 100,000, Jul 2022 - Jun 2023**)
- (**NSF, Faculty mentor**) REU Site: Interdisciplinary Computational Biology (**$ 413,526, Mar 2022 - Feb 2025**)
- (**SimCenter, Co-PI**) Addressing sampling biases in genome-wide association study for SAR-CoV-2 (**$ 90,000, Jul 2021 - Jun 2022**)
- (**NIH, Co-I**) A computational modeling framework for COVID-19 vaccination (**$ 122,580, Jun 2021 - May 2022**)

Teaching and Advising

Teaching at UTC

- MATH 2100: Introductory Statistics (one or two sessions every semester since Fall 2020)
- MATH 4130: Introduction to Probability and Statistics (Fall 2021, 2022)
- MATH 4140: Mathematical Statistics (Spring 2022, 2023,2024)
- MATH 4170: Nonparametric Statistics (Spring 2021)
- MATH 5130: Introduction to Probability and Statistics (Fall 2021, 2023,2024, 2025)
- MATH 5140: Mathematical Statistics Spring (2022, 2023)
- MATH 5150: Introduction to Biostatistics (Fall 2022)
- MATH 5170: Nonparametric Statistics (Spring 2021)
- MATH 5180: Analysis of Variance and Design of Experiments (Fall 2023)

Teaching at NMSU

- All math courses, eligible for graduate students teaching, were taught at least once, such as College Algebra, Intermediate Algebra, Pre-Calculus and Trigonometry, Calculus I, Calculus II, Statistics for Business and Behavior Sciences, Probability and Statistics for Engineers

Advising Students

Graduate Committee

- Chathuri Aththanayake (Ph.D., graduated in Summer 2025)
- Opoku Foster (Master, graduated in Spring 2025)
- Megan McCoy (Ph.D., in progress)
- Gertrude Osei (Master, graduated in Summer 2023)

Recommender

- Wrote recommendation letters for undergraduates, graduates to graduate schools and scholarships

Students Mentored

- (Spring 2025) Mentored graduate student (Benedicta Puplampu)
- (Spring 2025) Mentored undergraduate student (Reid Wilson)
- (Fall 2024) Mentored graduate student (Benedicta Puplampu)
- (Summer 2024) Mentored graduate students (Foster Opoku and Hagar Cobbinah)
- (Spring 2024) Mentored graduate student (Foster Opoku)
- (Summer 2023) Mentored graduate student (Andrew Stuckey)
- (Spring 2023) Mentored graduate student (Gertrude Osei)
- (Summer 2022) Mentored undergraduates (Vanessa Ramos from Jarvis Christian College and Will Torres from Adelphi University) in REU
- (Summer 2021) Mentored undergraduates (Landon Kehr from Taylor University and Bosah Daniel from Bronx Community College of the City University of New York) in REU

Invited Conference/Workshop Presentations

- **NMSU Math Colloquium Talk** at New Mexico State University in April 2025
- **Alamo Symposium in Statistics** at The University of Texas at San Antonio in March 2023
- **International Forum on Collaborative Development of Machine Learning and Digital Economy (virtual)** at Xi'an Technological University, Xi'an China in Dec. 2022
- **AMS Southeastern Sectional Meeting** at UTC in Oct. 2022
- **New England Statistics Symposium (virtual)** at the University of Connecticut in May 2022
- **Joint NMSU/UTEP Workshop** at NMSU or UTEP 11 times from Fall 2014 to Fall 2019

Contributed Talks

- **JMM 2020** Joint Mathematics MeetingsInference on the difference of location parameters under Multivariate skew-normal setting. Denver, CO

- **GRAS 2019** Graduate Research & Arts Symposium (GRAS)The plausibility regions for shape parameters under multivariate skew normal settings based on Inferential Model. NMSU, Las Cruces, NM
- **JRC 2018** Joint Research Conference on Statistics in Quality, Industry, and TechnologyInferences on the location parameter under multivariate skew normal settings. Sante Fe, NM
- **JMM 2018** Joint Mathematics MeetingInferences on the location parameter under multivariate skew normal settings. San Degio, CA

Poster Presentation

- **2025 Southeastern Surgical Congress SESC Annual Meeting** Re-evaluating the Necessity of Empiric Anti Fungal Coverage in Gastroduodenal Perforations **Presenter: Raine J.** New Orleans, LA
- **2024 American College of Chest Physicians CHEST Annual Meeting** Association of Heterozygotes for the Alpha-1-antitrypsin Gene With Cirrhosis and Mortality **Presenter: Patrick Koo.** Boston, MA
- **2024 Southern Medical Association Virtual Abstract Competition** Association of A1AT Heterozygotes with Pulmonary Fibrosis and Mortality **Presenter: Isaac Hembree.** Selected in the Top 10 Poster at the Annual Scientific Assembly
- **2024 American Thoracic Society International Conference** Effect of Heterozygote Alpha-1-antitrypsin Gene on the Development of Pulmonary Fibrosis and Mortality. **Presenter: Nathan Brott.** San Diego, CA
- **2024 UTC Fall Research Conference** Risk Analysis of A1at Heterozygotes with Pulmonary Hospitalization, Readmission, Fibrosis, And Mortality. **Presenter: Hagar Cobbinah.** UTC
- **2024 The 9th Workshop on Biostatistics and Bioinformatics** Prevalence of PI*Z and PI*S alleles of alpha-1-antitrypsin deficiency in Tennessee: A Study on Hardy-Weinberg Equilibrium. **Presenter: Foster Opoku.** Atlanta, GA
- **2024 UTC Spring Research and Arts Conference** Prevalence of PI*Z and PI*S alleles of alpha-1-antitrypsin deficiency in Tennessee: A Study on Hardy-Weinberg Equilibrium. **Presenter: Foster Opoku.** UTC
- **JSM 2019** Joint Statistical MeetingsThe decomposition of quadratic forms under matrix variate skew normal Distribution. Denver, CO

Seminar

- **2025 UTC Mathematics Colloquium Series** Some work on "wrong skewness" problem in stochastic frontier model. UTC
- **2024 UTC Mathematics Colloquium Series** The development and application of generalized F distribution. UTC
- **2023 UTC Mathematics Colloquium Series** Estimation for Skew-normal based stochastic frontier model. UTC
- **2022 NMSU statistics group Seminar (Zoom)** Inferences on Location Parameters Based on Independent Multivariate Skew Normal Distributions. NMSU
- **2022 UTC Mathematics Colloquium Series** An application of Support Vector Machine (SVM) in Arrhythmia Heartbeats Classification. UTC
- **2021 UTC Mathematics Colloquium Series** A Stochastic Gompertz Diffusion Model for Untreated Human Glioblastomas. UTC
- **2021 NMSU statistics group Seminar (Zoom)** Estimation of Parameters and Decomposition of Quadratic Forms Under Skew-normal Settings. NMSU
- **2019 NMSU Machine Learning Seminar** Prototype Methods and Nearest-Neighbors. NMSU
- **2018 NMSU statistics group Seminar** Skew normality test. NMSU
- **2018 NMSU Bio-math Seminar** Summer research presentation, Building an SDE model for breast cancer. NMSU

- **2017 NMSU Bio-math Seminar** Stochastic Differential Equations – Stochastic growth pattern of untreated human glioblastomas predicts the optimal time for surgery. NMSU
- **2016 NMSU statistics group Seminar** Skew symmetric distributions. NMSU

Honors and Scholarship

- **NSF QIP Mathematical Fellowship 2025** Partially supporting travel to attend QIP 2025
- **UTC Math Travel Grant** Partially supporting travel to attend QIP 2025
- **NSF SIAM Quantum Intersections Convening, Oct. 2024** Supporting travel to attend the workshop
- **UTC Math Travel Grant** Supporting travel for Alamo Symposium in Statistics at The University of Texas at San Antonio in March 2023
- **Dean's Graduate Award for Excellence of College of Arts and Sciences, Dec. 2019** The highest award given to a graduate student from the College of Arts and Sciences at NMSU
- **Outstanding Graduate Student 2018 -2019** at NMSU
- **ASA Joint Research Conference Student Travel Award 2018** at Santa Fe, NM
- **ASA Joint Research Conference Student Scholarship 2018** at Santa Fe, NM
- **Graduate Assistant Tuition Fellowship 2015 - 2018** at NMSU

Service

Committees

- University
  - **Committee member** in Course learning evaluations (2024 - 2025)
  - **Member** in Faculty Senate (2023- 2025)
- College
  - **Committee member** in College of Arts & Sciences Curriculum Committee (2022 - 2024)
- Department
  - **Committee Chair**
    Academic Awards Committee (2024 - 2027)
  - **Committee member**
    Undergraduate Committee (2024 - 2025)
    Advisory Committee (2023- 2024, 2026 - 2027)
    Graduate Committee (2023- 20257)
    Budget Committee (2022 - 2024)
    Academic Awards Committee (2021 - 2024)
    Math Competitions Committee (2021 - 2022)

Organizer/Reviewer

- **Co-organizer BCBST Actuarial Fall Visit** in Fall 2023
- **Session Chair** in International Forum on Collaborative Development of Machine Learning and Digital Economy (virtual) at Xi'an Technological University, Xi'an China in Dec. 2022
- **Co-organizer** for AMS Southeastern Sectional Meeting at UTC in Oct. 2022
- **Reviewer** for 30 journals within my expertise and ACM Mid-Southeast Conference 2021

Volunteers

- **Volunteer monitor** for 2025 JSM Continuing Education course
- **Volunteer proctor** for 2024 Math Kangaroo
- **Volunteer grader** for 2024 TMTA Math Competition
- **Volunteer** for 2024 TVR Middle School Drone Competition Championship
- **Volunteer Judge for MAA Student Poster Session - Applied Mathematics Session** during Joint Mathematics Meetings 2021